



HK IMO Training 2023–2024
Problem Set 13
For 13 January 2024



Problem 49. All the positive integers are coloured in 3 colours randomly. Prove that there are two positive integers of the same colour such that their difference is a perfect square.

Problem 50. Let Ω be the circumcircle of $\triangle ABC$. Let ℓ_1 and ℓ_2 be parallel straight lines passing through B and C respectively. Suppose ℓ_1 and ℓ_2 meet Ω at D and E respectively, such that both D and E lie on the same side of the line BC as A . The lines AD and CE meet at P . The lines AE and BD meet at Q . Let O, O_1, O_2, O' be the circumcentres of $\triangle ABC, \triangle ADQ, \triangle AEP$ and $\triangle OO_1O_2$ respectively. Prove that $OO' \parallel \ell_1$.

Problem 51. Define a sequence (x_n) by $x_1 = \sqrt{2}$ and

$$x_{n+1} = x_n + \frac{1}{x_n}$$

for all $n \geq 1$. Prove that

$$\sum_{j=1}^{2019} \frac{x_j^2}{2x_j x_{j+1} - 1} > \frac{2019^2}{x_{2019}^2 + x_{2019}^{-2}}.$$

Problem 52. Let n be a positive integer such that $n \equiv 2 \pmod{4}$. For each prime number p , let a_p be the arithmetic mean of the numbers $\left\{\frac{1^n}{p}\right\}, \left\{\frac{2^n}{p}\right\}, \dots, \left\{\frac{\left(\frac{p-1}{2}\right)^n}{p}\right\}$. Prove that there are infinitely many primes p with the same value of a_p .